Orthonormality / normal Matrices

$$\begin{bmatrix}1\\0\\0\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} \longrightarrow \text{perpendicular.}$$

$$\begin{bmatrix}a\\b\\c\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix}$$

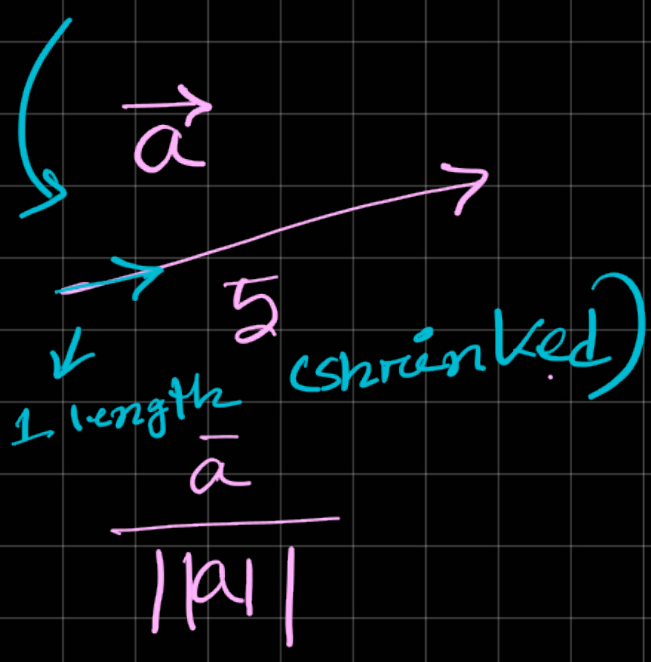
$$a^T b = 0$$

orthogonal → vectors → vector space → Matrix

$$M = \begin{bmatrix} | & | & & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & & | \end{bmatrix} \rightarrow \text{would be orthogonal if}$$

all the cols are orthogonal to 1 another. Dot product should give zero.

Normalize 0 to 1 ?



$$\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \rightarrow \text{Normalize} \Rightarrow \text{Length}$$

$$= \sqrt{1^2 + 2^2 + (-2)^2}$$

$$= 3$$

$$\therefore \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix} \text{ Normalized}$$

$$M = \begin{bmatrix} | & | & & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & & | \end{bmatrix}$$

orthonormal * (only if it $n \times n$ (sq matrix))

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix} =$$

↳ rotates any vector in space by

θ^0.

$$\begin{bmatrix} 1/\sqrt{6} & 0 \\ 1/\sqrt{6} & -2/\sqrt{5} \\ 2/\sqrt{6} & 1/\sqrt{5} \end{bmatrix}$$

→ Normalized now.

Not orthonormal Matrix as not sq.

Dot product 0. so orthonormal vector.

if I take 2 col from Q,

$$q_i^T q_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

→ all perpendicular.

$a^T a = (\text{length})^2 =$ sq. of Magnitude

$$Q^T Q = \begin{bmatrix} — & q_1 & — \\ — & q_2 & — \\ & \vdots & \end{bmatrix} \begin{bmatrix} | & | & | \\ q_1 & q_2 & \cdots \\ | & | & | \end{bmatrix}$$

$$\begin{bmatrix} - & q_n & - \end{bmatrix} \begin{bmatrix} & & \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \cdots \\ 0 & 1 & \cdots \\ 0 & \cdots & 1 \\ \vdots & \cdots & \cdots & 1 \end{bmatrix}$$

any matrix A^TA = symmetric

$$Q^TQ = I \Rightarrow Q^TQQ^{-1} = Q^{-1}I$$

$$\Rightarrow Q^T = Q^{-1}$$

° if Q was a sq mat,

$$Q^{-1} = Q^T$$ [if not sq, inverse doesn't exist].

perpendicular cols are always independent. So spans the whole space.

project vector b onto C(A).

$Pb = p$ → don't want

project vector b onto C(Q).

$$P = A(A^TA)^{-1}A^T$$

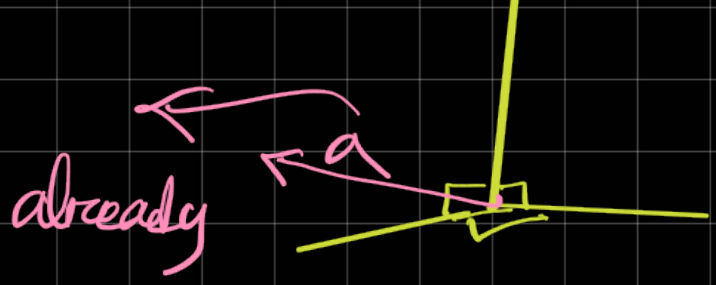
$$\Rightarrow P = Q(Q^TQ)^{-1}Q^T$$

$$= Q(I)^{-1}Q^T$$

$$= QQ^T$$

$Q^TQ = QQ^T$ [only when Q is sq]

$$\begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix}$$

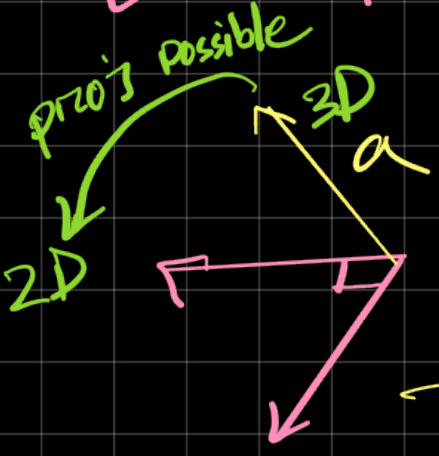


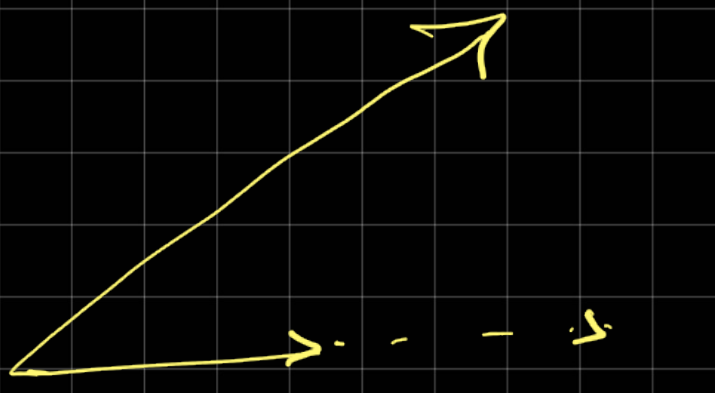
→ C(A) = All of 3D

in 3D space
no need to project.

$$\begin{bmatrix} d_1 & d_2 \\ 1 & 1 \end{bmatrix}$$

C(A) = A plane





→ the projection of a is not identity

$Ax = b$ → 99% won't be ✗

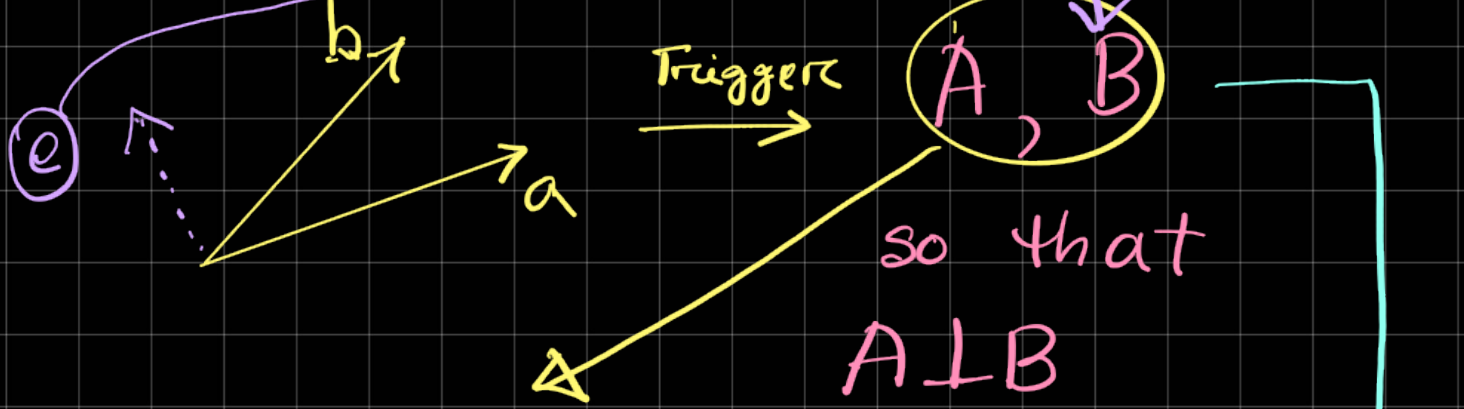
$A^T A\hat{x} = A^T b$ ✓✓ 99% this case exists.

$Q^T Q\hat{x} = Q^T b$ (I)

$$\hat{x} = Q^T b$$

$$\|v+w\| \leq \|v\| + \|w\|$$

Gram Schimdt is actually the mythical B changed to



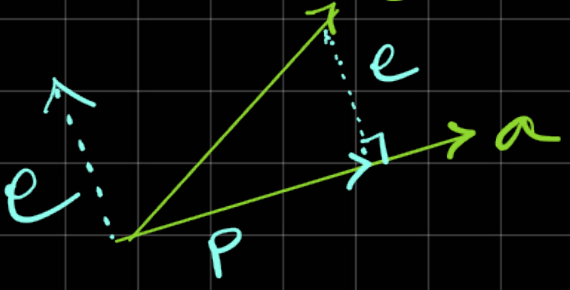
☆ Vectors Not Matrix.

$$\frac{A}{\|A\|}, \frac{B}{\|B\|}$$

q_1 q_2

Keep a unchanged and change b to make angle 90°.

b



$$b = p + e$$

$$\Rightarrow e = b - p = b - \left(\frac{A^T b}{A^T A} A \right)$$ done ✓

all vectors

projection of a vector on another vector.

To figure a vector that is perpendicular to other 2 or more vectors.
a, b, ©

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minus projection on top of a
" " " " " b
} subtract from both side

so we get component that is perpendicular along both a & b.

$$e = c - \left(\frac{A^T c}{A^T A} A \right) - \left(\frac{B^T c}{B^T B} B \right)$$

$C = C = \left(A^T A\right) \left(B^T B\right)$

or cross product of a b.

<u>practice</u> : $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 5 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 1 \end{bmatrix}$

Apply Gram-S to convert to

q_1, q_2, q_3.

$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$